MILITARY SPECIFICATION

SYRINGE, DENTAL RESTORATIVE MATERIAL, AND ACCESSORY KIT

This specification is approved for use by the Defense Logistics Agency-DM, Department of Defense, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a dental syringe and accessory kit of replacement components suitable for use in extruding composite restorative materials.

1.2 Classification. The syringe with replacement components shall be of the following types, as specified (see 6.1):

Type I - Syringe, complete (with replacement tips and plugs).
Type II - Accessory kit (replacement tips and plugs).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical, General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions), any any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.
FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24344 - SYRINGE AND ACCESSORY KIT.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2240 Standard Test Method for Rubber Property-Durometer Hardness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. The materials, colors, finishes and hardnesses of the components of the syringe, tips and plugs shall be as specified in table I. Materials and properties not specified shall be entirely suitable for the intended purpose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

TABLE I. Materials, colors, finishes and hardnesses.

Component	Material	Color	Finish	Hardness
Syringe body				
Design No. 1	Stainless steel (AISI 303 except flange shall be AISI 304)		Machine finish inside; satin finish outside	
Design No. 2	Plastic, rigid (styrene - acrylonitrile)	Light blue (or light color)	Smooth	
Plunger	Nylon, rigid	Black or dark color)	Smooth	
Plugs	Neoprene synthetic rubber, prelubricated	Gray (or light color)	Smooth	60 ± 5 International hardness degrees
Tips	Polypropylene, semi-rigid	Transparent, colorless (clear or natural color)	Smooth	

3.2 Style, design and dimensions. Style, design, and dimensions of the complete syringe with replacement parts shall be in conformance with Defense Personnel Support Center (DPSC) Drawing 24344.

3.2.1 Type I, Syringe, complete (with tips and plugs). Type I, complete syringe (with tips and plugs) shall be suitable for use in extruding composite restorative materials or cements using the disposable tip and plug. The syringe shall consist of a syringe body, plunger, angulated conical tip, and conical plug. The syringe body shall have a flanged finger support having anti-roll flats, and shall house a semi-captive plunger. Plunger diameter shall be slightly larger near the distal end (conical tip end), and the syringe body diameter shall be restricted at the proximal end (flange end) to retain the plunger in the body. The plunger shall be readily removable by hand upon exertion of axial pressure (pull) not exceeding 5 pounds, when tested as specified in 4.5.3.

3.2.1.1 Syringe body construction. Syringe body shall be furnished in stainless steel (Design No. 1) or plastic (Design No. 2). The stainless steel syringe body shall be of two piece construction (barrel and flange). The flange shall be permanently secured to the barrel. The plastic syringe body shall be of one piece molded construction.

3.2.2 Type II Accessory kit. Type II accessory kit, replacement tips and plugs, shall be suitable for use with type I syringe in placing composite restorative materials or cements during dental procedures. Tips shall have a shoulder at the proximal end (back end) for retention in the syringe, and an angulated conical tip with a small orifice at the distal end (pointed end) for extrusion of material. It shall be possible to bend the conical tips by hand to facilitate placement in difficult to reach areas. Plugs shall be approximately conical in shape having a blunted apex.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

3.3 Corrosion resistance. The stainless steel components shall meet the requirements of GG-1-526, when tested as specified in 4.5.2.

3.4 Hardness. The neoprene shall have the hardness specified in table I, when tested as specified in 4.5.4.

3.5 Finish. Finishes of components shall be as specified in table I. Mating surfaces of all components shall be smoothly finished for ease of operation.

3.6 Marking. The type I syringe body shall be permanently and legibly marked with the name or registered trademark of the contractor. Syringes of foreign manufacture shall also be marked with the name of the country of origin.

3.7 Instructions. Adequate and accurate instructions for assembly and use of the syringe shall be included with each type I complete syringe (with tips and plugs).

3.8 Workmanship. Workmanship shall be first class throughout. The complete syringe (type I) and replacement components (type II) shall be supplied clean and free from defects which may detract from their appearance or impair their serviceability. The components of the syringe shall fit and mate with each other in such a way that no leakage or clogging shall occur during the extrusion of material.

3.9 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval from the contracting officer.

3.10 Quantity. Type I, syringe (with tips and plugs) shall consist of a package containing the assembled syringe (body and plunger) and a minimum of 50 disposable tips in a package and 50 disposable plugs in a separate package. Type II, accessory kit shall consist of a package containing a minimum of 100 disposable tips in a package and 100 disposable plugs in a separate package. The accessory kit (tips and plugs) furnished in a package shall be identical in construction to the tips and plugs furnished with type I complete syringe.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials used in the syringe, tips, and plugs, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and IV (see 4.2.1 and 4.3) and table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. The unit of product for examination purposes shall be one type I syringe or one type II accessory kit, as applicable. A minimum of 3 tips and 3 plugs from a unit of issue of either type I syringe or type II kit (as applicable) shall be taken for examination purposes.

TABLE II. Examination criteria.

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

TABLE III. Test criteria.

Characteristic	Requirement	Test procedure	Unit of product	Inspection level
Chemical analysis (steel component)	3.1	4.5.1	Not applicable	1/
Corrosion resistance (steel component)	3.3	4.5.2	One syringe (type I)	S-2
Hardness (rubber component)	3.4	4.5.4	Not applicable	1/
Plunger removal	3.2.1	4.5.3	One syringe (type I)	S-2

1/ A sufficient quantity from each lot or batch of stainless steel or rubber (as applicable) used shall be tested.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The syringe and replacement components shall be examined for defects including, but not limited to, those listed in table IV.

TABLE IV. Classification of defects. 1/

Categories	Defects
Major A:	
101	Type not as specified.
102	Components of syringe fail to fit or function properly.
103	Style, design or construction of any component of syringe assembly not as specified.
Major B:	
151	Dimensions affecting fit of components not as specified.
Minor A:	
201	Finish not as specified.
202	Disposable tips not semi-rigid or colorless or transparent (clear or natural color).
203	Plunger not rigid or black (or dark color).
204	Plugs not gray (or other light color).
205	Plastic syringe body not light blue (or other light color).
206	Instructions for proper assembly, use, and cleaning of the syringe either missing, inadequate, inaccurate, or illegible.
207	Identification marking on syringe body missing, inadequate, inaccurate, or illegible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects 1/ - Continued

Categories	Defects
Minor B:	
251	Any dimension (except 151) not as specified.
252	Quantities of tips or plugs furnished not as specified for type I or II.

1/ In process examination may be used in checking for listed defects. In examination for dimensions of the tip, manufacturer's molds may be measured in lieu of the tip itself.

4.4 Packaging inspection. The inspection of the unit packaging, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests.

4.5.1 Chemical analysis. In the absence of certificates of quality, chemical analysis of the steel syringe body shall be performed in accordance with method 111 or 112 of FED-STD-151. In the event of conflict, method 111 shall govern.

4.5.2 Corrosion resistance. The steel body of the syringe shall be subjected to the boil test, as specified in GG-I-526.

4.5.3 Plunger removal. The syringe shall be suspended by the distal end and a load of five pounds shall be applied to the thumb or palm rest portion of the plunger. The assembly shall be observed to see if the weight causes removal of the plunger.

4.5.4 Hardness test. The neoprene rubber used in the fabrication of the plugs shall be subjected to the durometer hardness test in accordance with method 3021 of FED-STD-601 or American Society for Testing and Materials (ASTM) method D2240. Specimens for tests shall be prepared simultaneously with manufacture of the rubber plugs in accordance with the requirements of section 2 of method 3021.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing one syringe along with fifty plastic tips and fifty rubber plugs (type I) or one package containing one hundred rubber plugs and one hundred plastic tips (type II), as specified, constitutes one unit of issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

5.1.2 Level A.

5.1.2.1 Unit packaging.

5.1.2.1.1 Type I. A minimum of fifty plugs and fifty tips shall be clean, dry and separately packaged in clear plastic or polyethylene bags of an appropriate size. Closure of the bags shall be effected by heat sealing or other suitable means to prevent spilling of contents. The packaged tips and plugs, along with an assembled syringe, shall be packaged in a box conforming to PPP-B-676, type I. Closure shall be as specified in the appendix of the box specification.

5.1.2.1.2 Type 2. A minimum of one hundred tips and one hundred plugs shall be clean, dry and separately packaged in clear plastic or polyethylene bags of an appropriate size. Closure of the bags shall be effected by heat sealing or other suitable means to prevent spilling of contents. The bags containing the plugs and tips shall be further packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Twelve unit packages containing type I, syringe complete with plugs and tips, or twelve unit packages containing type II, accessory kit (tips and plugs), shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Each type I or type II unit of issue shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Six intermediate packages (72 units) containing type I, syringe complete with plugs and tips, or eight intermediate packages (96 units) containing type II, accessory kit (tips and plugs), shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Six intermediate packages (72 units) containing type I, syringe complete with plugs and tips, or eight intermediate packages (96 units) containing type II, accessory kit (tips and plugs), shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged type I, syringe complete with plugs and tips or the packaged type II, accessory kit (tips and plugs), shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Interior packages. Each unit and intermediate package shall be marked as specified in MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation
6520-00-139-4580	ACCESSORY KIT, DENTAL FILLING MATERIAL SYRINGE	Type II
6520-00-139-4583	SYRINGE, DENTAL FILLING MATERIAL, Thumb Rest Handle	Type I

6.3 Supersession data. This specification includes the requirements of Purchase Description No. 2 dated 4 October 1974.

Preparing activity:
DLA-DM

Project No. 6520-1467

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-37846(DLA-DM) SYRINGE, DENTAL RESTORATIVE MATERIAL, AND ACCESSORY KIT

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:37Z
Check the source to verify that this is the current version before use.